

Review

A Comprehensive Review of Deep-Learning Applications to Power Quality Analysis

Indu Sekhar Samanta ¹, Subhasis Panda ², Pravat Kumar Rout ³, Mohit Bajaj ^{4,5,6,*}, Marian Piecha ⁷,
Vojtech Blazek ⁸ and Lukas Prokop ^{8,*}

- ¹ Department of Computer Science Engineering, Siksha 'O' Anusandhan University, Odisha 751030, India
² Department of Electrical Engineering, Siksha 'O' Anusandhan University, Odisha 751030, India
³ Department of Electrical and Electronics Engineering, Siksha 'O' Anusandhan University, Odisha 751030, India
⁴ Department of Electrical Engineering, Graphic Era (Deemed to be University), Dehradun 248002, India
⁵ Graphic Era Hill University, Dehradun 248002, India
⁶ Applied Science Research Center, Applied Science Private University, Amman 11937, Jordan
⁷ Ministry of Industry and Trade, 11015 Prague, Czech Republic
⁸ ENET Centre, VSB—Technical University of Ostrava, 70800 Ostrava, Czech Republic
* Correspondence: thebestbajaj@gmail.com (M.B.); lukas.prokop@vsb.cz (L.P.)

Abstract: Power quality (PQ) monitoring and detection has emerged as an essential requirement due to the proliferation of sensitive power electronic interfacing devices, electric vehicle charging stations, energy storage devices, and distributed generation energy sources in the recent smart grid and microgrid scenarios. Even though, to date, the traditional approaches play a vital role in providing a solution to the above issue, the limitations, such as the requirement of significant human effort and not being scalable for large-scale power systems, force us to think of alternative approaches. Looking at a better perspective, deep-learning (DL) has gained the main attraction for various researchers due to its inherent capability to classify the data by extracting dominating and prominent features. This manuscript attempts to provide a comprehensive review of PQ detection and classification based on DL approaches to explore its potential, efficiency, and consistency to produce results accurately. In addition, this state-of-the-art review offers an overview of the novel concepts and the step-by-step method for detecting and classifying PQ events. This review has been presented categorically with DL approaches, such as convolutional neural networks (CNNs), autoencoders, and recurrent neural networks (RNNs), to analyze PQ data. This paper also highlights the challenges and limitations of using DL for PQ analysis, and identifies potential areas for future research. This review concludes that DL algorithms have shown promising PQ detection and classification results, and could replace traditional methods.

Keywords: deep-learning (DL); machine learning (ML); artificial intelligence (AI); power quality monitoring and detection; feature extraction; classification of PQ disturbance; artificial neural network (ANN)



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1. Introduction

With many inherent and associated issues in power systems, particularly in smart grids and microgrid environments, PQ disturbance monitoring and detection is indispensable for secure, reliable, well-controlled, and protective system operation [1]. An advanced monitoring system with proper techniques to analyze PQ events is required to meet the consumer's requirements for a quality power supply. This is due to the adverse effect of poor PQ, such as equipment damage, production losses, and safety hazards. In addition, the integration of power electronics-based interfacing devices, nonlinear loads, EV charging, renewable energy integration, and energy storage devices take the PQ issues beyond control, and make it difficult to maintain within a limit [2–5]. Secondly, accurately and quickly

detecting PQ events is necessary to take further corrective action leading to a stable and protective energy system. Remedial measures must be adopted in two aspects: using better filters and compensating devices or monitoring, and detecting through proper devices and techniques [6]. Therefore, accurate and quick detection of PQ events is critical and has a vital influence on the energy supply's reliability, security, and quality. Fortunately, many innovative and emerging technologies have been applied to the smart grid to facilitate real-time PQ event detection and identification. DL has recently gained considerable attention and significant advancements, with numerous applications in various engineering and technology fields, including power systems [7]. This review has been conducted with the motivation that DL can be a promising and better technique for PQ detection and identification in the future. The primary objective of this comprehensive review is to present the current state-of-the-art research, and identify potential research directions for a better solution based on DL-based approaches.

Generally, a two-step process has been followed for PQ detection and classification. The features are extracted from the nonstationary power signals for various PQ disturbances in the first step. These features are used to train the ML classifiers according to the characteristics of respective PQ event signals, and classify them for better identification to take further decisions and countermeasures accordingly. One of the major advantages of DL is its capability to extract features during the training period [6]. The feature extraction process in DL is fully automated, which is why it leads to more accurate results. In many DL applications, feature extraction and classifier learning are performed simultaneously, making it an end-to-end learning system. DL can be applied directly to perform the classification task, and can accomplish state-of-the-art accuracy with high performance [8]. DL is considered a branch of ML, and is applied to learn in those applications where high-level abstractions are required from data. DL tries to learn high-level features from PQ data incrementally; this ability eliminates the need for domain expertise and hardcore feature extraction [9]. Due to this, all of the DL-based models are presented here, focusing on PQ analysis, and intentionally avoiding discussing feature extraction and classification separately. These characteristics of DL attract researchers to apply them to various PQ analyses. The above is the primary motivation to present the DL application for PQ event detection in detail in this review.

Automatic feature extraction is one of the major causes of attraction of DL compared to other ML techniques. Recently, DL has emerged as an efficient technique to extract the prominent and dominant features to characterize the nonstationary power signals for various applications, particularly for PQ event detection and identification. Firstly, the DL algorithm can generate new features, even with a limited number of samples available as the training dataset, and perform a complex task. Secondly, DL can perform consistently, even with unstructured, overlapping, and mixed hybrid datasets in simultaneously occurring PQ events. Thirdly, DL can efficiently learn complex features and perform more intense computational tasks due to its multilayered structure. The above case is due to three major factors. The prominent factors are (1) the ability to learn from its errors, (2) verifying the accuracy of its predictions/outputs, and (3) making the necessary adjustments accordingly [7]. In addition to the above, it is found that, generally, DL performance increases proportionally to the training dataset's volume. This nature makes DL more feasible and better for large datasets under real-time conditions. Lastly, unlike the general neural network (NN) models, the DL model takes less time and is faster during training. Even though signal processing techniques are very efficient for feature selection for PQ event detection, few suggestions are made for DL for the same purpose [7]. DL can support multilabel classification problems to provide an accurate classification, which is much needed for the classification of a variety of PQ events. The DL models can easily define and evaluate multilabel classification tasks. In DL models, it learns to perform classification tasks directly from the PQ event datasets or the features, and can achieve state-of-the-art accuracy. The above factor indicates a wide possibility to further explore the application of DL in PQ event detection and classification. Looking at DL application at its infancy stage

in smart grid applications, particularly for the PQ detection analysis, this review motivates us to present an extensive study on it to reflect the concepts, design, models, and future scope of research.

DL has shown great potential for PQ detection and classification, since it can automatically learn complex features from the available raw data [7]. Numerous DL architectures, such as long short-term memory (LSTM) networks, generative adversarial networks (GANs), and convolutional neural networks (CNNs), have been applied to PQ analysis, achieving promising results. The primary work performed according to the objectives of this review are as follows.

- Systematically review recent research works that employed DL approaches for PQ detection and classification. It also presents their performance gain discussion, innovations made in formulation, and respective pros and cons.
- Analysis has been performed on the performance of various DL architectures on different PQ datasets, and discusses the advantages and limitations of the individual approach.
- Present the challenges and opportunities for future research to develop robust DL models for handling imbalanced and noisy datasets, and integrating DL-based PQ analysis into real-time monitoring and control systems.
- Provide a comprehensive overview of the up-to-date state-of-the-art AI-based PQ detection and classification, particularly using DL approaches to aid researchers, engineers, and practitioners in the power industry in designing more efficient and reliable power systems.
- Highlight the potential of DL-based PQ analysis for enabling new applications in smart grid systems.

To the best of our information, this is the first review paper on DL that sheds light on its application to PQ detection and analysis in the smart grid domain. By accurately detecting and classifying PQ disturbances in real-time, it is possible to predict the future behavior of electrical equipment and prevent failures before they occur. That is why it is necessary for a better method, and looking at all the perspectives of DL, this review has been conducted to find the research gaps and future scope. This review is helpful for all researchers to find a better technique based on the DL method for PQ disturbance classification in the smart grid environment. The remaining portion of the manuscript is planned and structured as follows. The workflow of DL for PQ event detection and identification has been presented in Section 2. Section 3 discusses the various DL techniques used for PQ analysis. Critical findings are analyzed in Section 4. The future scope referring to the research gaps is presented in Section 5. At last, the concluding remarks for the entire review are presented in Section 6.

2. Workflow of Deep-Learning for PQE Classification

The workflow for DL-based PQ event classification typically involves several stages, as shown in Figure 1.

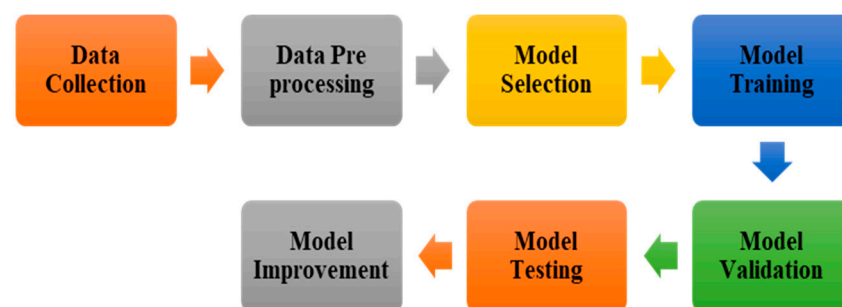


Figure 1. Workflow of deep-learning for PQE Classification.

Here is a general overview of the workflow:

Step-1 data collection: this is the first step in any ML-based process to collect data. Regarding the PQ event classification, data can be collected from various sources, such as phasor measurement units (PMUs), smart meters, and power quality monitors. The nature of data in the PQ analysis is generally nonstationary and in a discrete time-series form of voltage or current waveforms. It comprises information on the PQ event's type and duration.

Step-2 data preprocessing: the collected data may contain noise, missing values, or other inconsistencies. Therefore, it is essential to preprocess the data before training the model. This step may include many preprocessing stages, such as data cleaning, normalization, resampling, and feature extraction.

Step-3 model selection: the next step is to select a suitable DL model, which depends on the nature of the task to perform with the associated datasets. Several models, such as recurrent neural networks (RNNs), convolutional neural networks (CNNs), and transformer-based models, can be used.

Step-4 model training: once the model is selected, the model is trained using the preprocessed PQ datasets. The training process involves feeding the model with the training input datasets and optimally adjusting the model's parameters to minimize the loss function. The loss function acts as an index during the training process and measures how well the model performs, and the training process aims to minimize it.

Step-5 model validation: the model's performance is evaluated on a separate validation dataset after the training. This step helps to avoid overfitting, where the model performs fine on the training data and fails to perform better on the new data. The validation dataset should represent the test data and not be used during model training.

Step-6 model testing: once the model is trained and validated, it is ready for deployment. The model can classify PQ events in real-time or batch mode. The model outputs either a probability distribution over the different PQ event classes, or a binary decision that may represent the presence or absence of a specific PQ event.

Step-7 model improvement: finally, the model's performance is improved by fine-tuning the model parameters, adding more training data, or using ensemble models. The model's performance should be monitored regularly to ensure it is still relevant and accurate.

3. Deep-Learning Techniques for PQ Classification

DL is a subfield of ML that uses an ANN with multiple hidden layers to model complex relationships between inputs and outputs. The following are some DL methods used for PQ disturbance/event classification.

3.1. Convolutional Neural Network (CNN)

A CNN is a type of ANN designed generally for image recognition and processing that is particularly effective for image classification and object recognition tasks. However, it is also substantially applied to PQ analysis. CNNs consist of multiple layers of interconnected nodes, where the nodes in each layer perform mathematical operations on the outputs from the previous layer, as depicted in Figure 2. The key feature of a CNN is the convolutional layer, where a small matrix (called a filter or kernel) is applied to local regions of the input data, producing a filtered output that is then passed on to the next layer [10]. The filters are designed to extract useful features from the input datasets, which are then processed in subsequent layers to perform object recognition. CNNs can also be trained end-to-end using supervised learning techniques, making them highly adaptable to various image processing and recognition tasks [10].

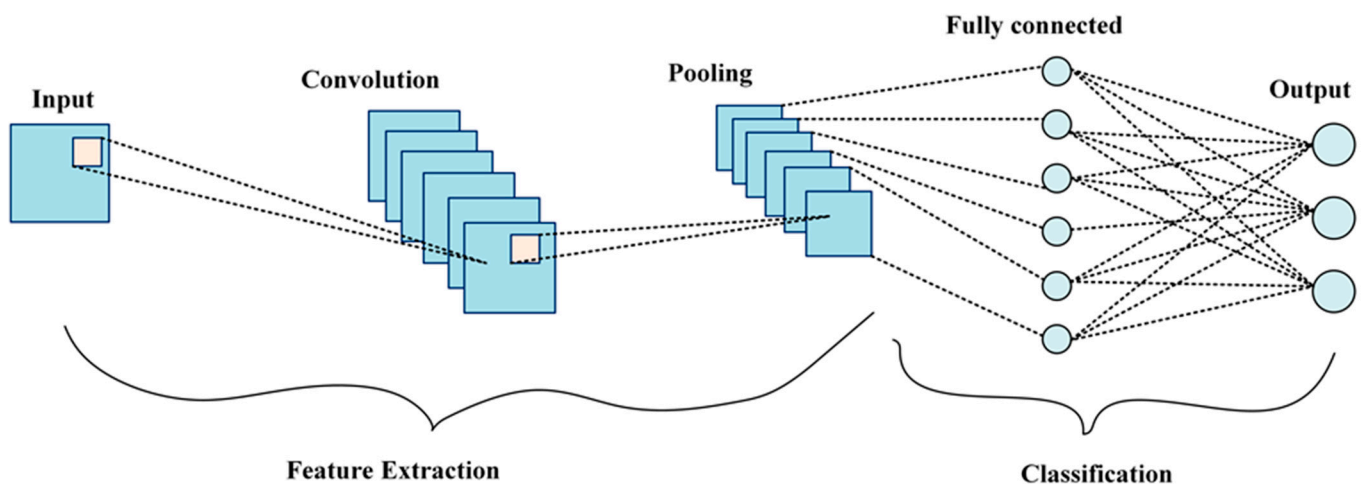


Figure 2. Architecture of Convolutional Neural Network.

The algorithm of a CNN for PQ detection and analysis comprises five major stages [11]. Stage-1 convolution stage: the input image is convolved with multiple filters to extract features. The result of the convolution is a feature map that encodes the information about the input image in a compact form. Stage-2 nonlinearity (ReLU): after that, the feature map is passed through a nonlinear activation function, such as a rectified linear unit (ReLU), which introduces nonlinearity into the network and allows it to learn more complex functions. Stage-3 pooling: the feature map is then downsampled using a pooling operation, such as max pooling, which reduces the spatial dimensions of the feature map while retaining the most important information. Stage-4 repeat: the process of convolution, nonlinearity, and pooling is repeated several times, with each iteration learning increasingly complex features. Stage-5 classification: lastly, the output of the last convolutional layer is fed into a fully connected layer, which performs the classification. The weights of the fully connected layer are optimally designed to be learned during training.

The CNN has many advantages, such as good performance on classification and object recognition tasks, robustness to translation and deformation, parameter sharing and pooling layers, reduced parameters and computational cost, and end-to-end training with supervised learning, which allow for high adaptability [10]. However, the limitations that need to be overcome with further innovative modeling improvement are the requirement of large amounts of training data, the high computational expense, and overfit if not properly regularized. In addition, it is very sensitive to the choice of hyperparameters, such as the number of filters, kernel size, and stride. The filters learned by CNNs can be difficult to interpret, making understanding how the network makes its decisions challenging.

Many authors suggested CNN-based approaches for PQ event classification. In recent years, CNNs have shown remarkable performance in PQ event classification due to their ability to automatically extract relevant features from time-series data. In this review, the performance and accuracy of CNN-based models used for PQ event classification are reported in the following research papers. Ekici et al. (2021) proposed an optimized Bayesian CNN for PQ event classification [12]. The results showed that the proposed method outperformed other traditional machine learning algorithms, such as decision trees and random forests, regarding accuracy. The overall accuracy achieved by the model was reported as 96.5%. The proposed model was evaluated on the publicly available PQ dataset. Ramalingappa and Manjunatha (2022) used a hybrid model with a complex wavelet phasor model and a customized CNN to classify PQ events [13]. The dataset used in this study was collected from the power grid in India. The proposed model achieved an accuracy of 99.33% in classifying power quality events. Mohan et al. (2017) proposed a DL architecture called Deep Power for PQ disturbances classification [14]. The proposed model was evaluated on the publicly available UCI Electric PQ dataset, and the results showed

that the model achieved an accuracy of 99.71%. Sahani and Dash (2020) proposed an FPGA-based deep CNN for PQ event recognition [15]. The model was evaluated on synthetic and experimental data collected from process-adaptive VMD data. The overall accuracy achieved by the proposed method was reported as 96.75%. Shen et al. (2019) proposed an improved PCA and CNN-based framework for power quality disturbance monitoring and classification in wind grid distribution systems [16]. The model was trained on a dataset collected from a real wind farm in China, and the results showed that the proposed method achieved an overall accuracy of 96.2%. Qiu et al. (2019) proposed a multifusion CNN-based automatic identification framework for complex PQ disturbances; the proposed method fused information from the PQ event's time-domain and frequency-domain features [17]. The model was evaluated on a dataset collected from a PQ monitoring system, and the results showed that the proposed method achieved an overall accuracy of 98.46%.

In summary, the results reported in these papers show that CNN-based models can achieve high accuracy in PQ event classification. The accuracy achieved by these models ranges from 96.2% to 99.71%, depending on the dataset and methodology used. The proposed methods in these papers can be applied in real-world power systems to improve the efficiency of fault detection and diagnosis.

The CNN has proven to be a very powerful tool for image classification tasks, including the classification of PQ events. As technology advances, the future scope of CNNs for PQ event classification looks promising. Here are some potential areas of growth and development that can be focused on in future research to come in these directions. Increasing accuracy: researchers continue fine-tuning CNN models to achieve higher accuracy in PQ event classification. The accuracy of a model depends on several factors, including the size and quality of the training dataset, the architecture of the model, and the hyperparameters used during training. Real-time detection: there is a growing need for real-time detection of PQ events to enable early warning systems and quick response times. Future research focuses on developing CNN models that can perform PQ event classification in real-time. Multimodal data: PQ events can be analyzed using different data types, including waveform data, frequency spectra, and spectrograms. Researchers explore using multimodal data for PQ event classification, which could potentially improve the accuracy of CNN models. Transfer learning: in this learning, a pretrained CNN model is used as a starting point for a new task. This concept leads to an effective technique for improving the accuracy of CNN models for PQ event classification. Future research focuses on developing transfer learning techniques for various PQ events. Edge computing: where the processing is performed locally on devices such as sensors and smart meters and is becoming increasingly important in PQ event classification. Future research focuses on developing CNN models that can be deployed on edge devices, allowing for real-time processing and analysis of PQ events. Overall, the CNN has a bright future in PQ event classification, and ongoing research and development continue to improve its performance and broaden its application scope.

3.2. Recurrent Neural Networks (RNNs)

RNNs are deep neural networks, as shown in Figure 3, designed to handle sequential data, such as time-series signals. In PQ disturbance/event classification, RNNs are generally applied to model the temporal dependencies in the input datasets, which can improve the accuracy of the classification. One of the main advantages of RNNs is that they can effectively capture the temporal dependencies in the input datasets, which can be critical for accurately classifying PQ disturbances/events. Additionally, RNNs can handle large amounts of sequential data, which can be challenging for other classifiers. However, one of the major limitations of RNNs is that they can be computationally intensive, requiring many parameters and computations.

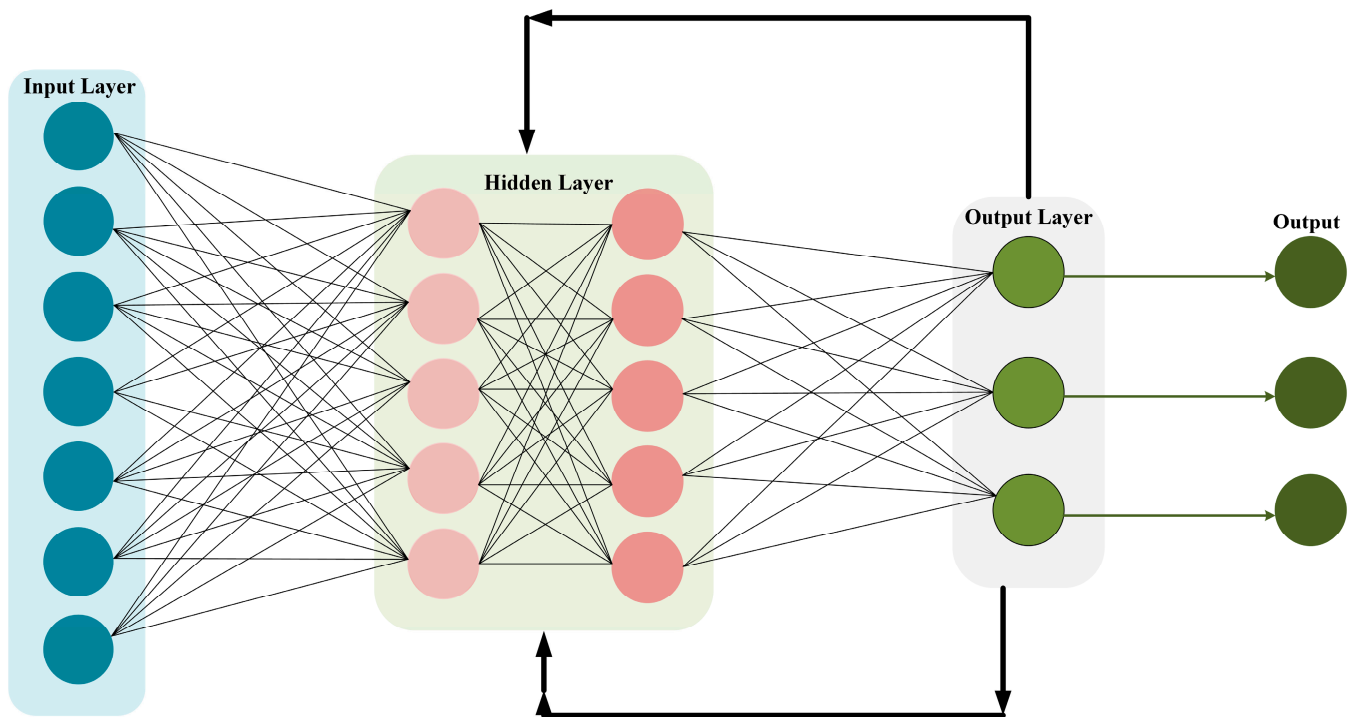


Figure 3. Architecture of recurrent neural networks.

The input data has been preprocessed by normalizing and transforming it into a suitable format for the RNN. The major steps of the algorithm are as follows [18]. Stage-1: initialize the network architecture and parameters. Stage-2: input the training data and labels. Stage-3: pass the input data through the hidden layers of the RNN. Stage-4: update the hidden state of the RNN at each time step. Stage-5: pass the hidden state through a fully connected layer for classification. Stage-6: train the network using backpropagation and gradient descent techniques. Stage-7: evaluate the performance of the network on the validation dataset. Stage-8: repeat steps 6–7 until the desired performance is achieved.

The major advantages of RNNs that are attractive to apply for PQ event detection and classification are as follows. The ability to process sequential data: RNNs can process input sequences of varying lengths and remember previous inputs through a hidden state, making them suitable for tasks. Flexibility: RNNs can be used in many different ways, such as feedforward or feedback form, making them highly adaptable to various tasks. Learning long-term dependencies: RNNs are designed to handle input sequences of arbitrary length and can learn long-term dependencies between sequence elements. Parallel processing: RNNs can be trained in parallel on different parts of a sequence, speeding up training and improving performance. However, RNNs have a few major limitations that must be focused on. Vanishing gradient problem [19]: RNNs can suffer from the vanishing gradient problem, where gradients become very small and cause the network to stop learning. The above is a common issue in networks with long-term dependencies. Computationally expensive: RNNs can be computationally expensive, especially when processing long sequences, making them less practical for real-time applications. Difficulty capturing global context [19]: RNNs typically process input sequences one element at a time, making it difficult to capture global context or long-range dependencies between distant elements in a sequence. Overfitting [20]: RNNs can be prone to overfitting, especially when working with small datasets, leading to poor generalization performance. Choosing the right architecture: choosing the right RNN architecture can be difficult, as there are many types of RNNs, each with its strengths and weaknesses.

PQ events have become a critical concern in power systems, as they can cause damage to sensitive electrical equipment and lead to significant economic losses. Various signal

processing and ML techniques have recently been employed to detect and classify PQ disturbances automatically. Among these techniques, RNNs have shown promising results due to their ability to capture the temporal dependencies in the power signal. Several research articles have explored the use of RNNs for PQ event classification. Lee et al. [21] proposed an online power quality disturbance classification method using an RNN. They used a dataset of synthetic power signals with various disturbances and achieved a classification accuracy of up to 99.9%. Mohan et al. [22] also used an RNN-based approach for PQ event classification, achieving up to 96.9% accuracy. Mishra [23] comprehensively reviewed PQ disturbance detection and classification methods using signal processing and soft computing techniques, including RNNs. Garcia et al. [24] compared the performance of three DL architectures, including RNNs, for PQ event classification and found that the CNN-LSTM model achieved the best accuracy of 98.97%. In a recent study, Dawood and Babulal [25] proposed a red deer-optimized RNN for PQ event classification and achieved an accuracy of 99.5%. Nagata et al. [26] developed a real-time voltage sag detection and classification method using an RNN and achieved an accuracy of up to 98.6%. Lee [27] also developed an automatic PQ monitoring system using an RNN. Aggarwal et al. [28] proposed a hybrid architecture combining wavelet transform, principal component analysis, and an RNN for PQ event classification and achieved an accuracy of up to 99.2%. Overall, RNNs have shown promising results in PQ event classification, with high accuracy levels ranging from 96.9% to 99.9%. These results indicate the potential of RNNs in developing reliable and efficient PQ monitoring systems for power systems.

RNNs have been widely used for classification tasks, including PQ events. RNNs are particularly useful for classifying PQ events because they can capture temporal dependencies in the data. RNNs can effectively model the temporal behavior of the signal. There are several future scopes for using RNNs to classify PQ events. Developing more advanced architectures: researchers can explore more advanced architectures, such as LSTM (long short-term memory) and GRU (gated recurrent unit), for better performance. These architectures can capture long-term dependencies and prevent the vanishing gradient problem in traditional RNNs. Handling high-dimensional data: PQ event classification often involves high-dimensional data from multiple sensors. Researchers can explore methods for handling such data, such as using attention mechanisms to focus on relevant parts of the data. Incorporating external factors: external factors such as weather, time of day, and location can affect PQ events. Researchers can explore methods for incorporating these factors into the classification model to improve accuracy. Developing transfer learning techniques: transfer learning can leverage pretrained models for classifying PQ events in new environments. The above learning reduces the labeled data required for training and improves the model's accuracy. Overall, the future of using RNNs to classify PQ events looks promising, and there are several avenues for further research and development.

3.3. Autoencoder (AE)

Autoencoders are deep neural networks, as shown in Figure 4, designed to learn compact representations of input datasets. In PQ disturbance/event classification, autoencoders can extract features from time-series signals, which can be input to other classifiers, such as SVM or k-NN. One of the main advantages of autoencoders is that they can effectively reduce the dimensionality of the input datasets, which can improve the computational efficiency of the classification process. Additionally, autoencoders can denoise the input data, improving classification accuracy. However, one of the major limitations of autoencoders is that they may be computationally intensive, as they require many parameters and computations.

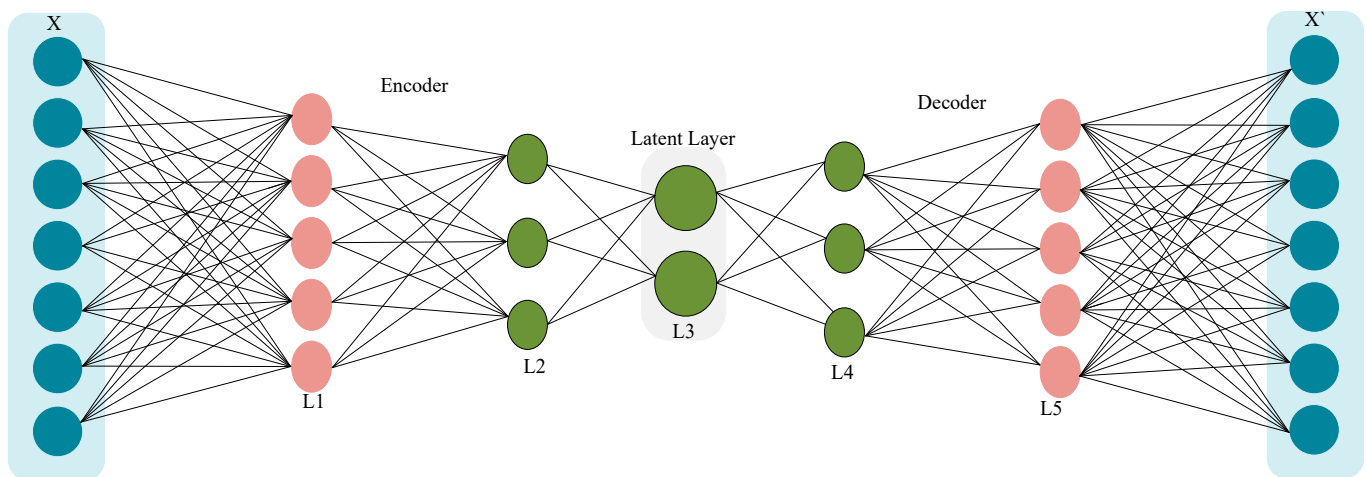


Figure 4. Architecture of autoencoder.

The major steps of the AE application are demonstrated as follows [29]. The process can be evaluated by following ten steps for the PQ analysis. Step-1: initialize the network architecture and parameters. Step-2: input the training data and labels. Step-3: pass the input datasets through the encoding layers of the autoencoder. Step-4: compress the input datasets into a lower-dimensional representation. Step-5: pass the compressed representation through the decoding layers of the autoencoder. Step-6: reconstruct the original input datasets from the compressed representation. Step-7: train the network using backpropagation and gradient descent techniques to minimize the reconstruction error. Step-8: extract the features from the encoding layer of the trained autoencoder. Step-9: use the extracted features as input to another classifier, such as SVM or k-NN. Step-10: evaluate the performance of the autoencoder-based classification system on the validation set.

The AE architecture is gaining popularity in PQ event detection and classification due to the following advantages [29]. **Dimensionality reduction:** autoencoders can learn an efficient representation of high-dimensional data, reducing the number of features required to represent it. The above factor can help to reduce the computational cost of training and enhance the model's generalization. **Unsupervised learning:** autoencoders can be trained on unlabeled data, which can be beneficial when labeled data are scarce or costly. The above advantage can be particularly useful for anomaly detection, where the model is trained to identify unusual patterns in the input datasets. **Nonlinear transformations:** autoencoders can learn complex, nonlinear transformations between the input and the latent representation. This transformation can help capture complex patterns in the data that may be difficult to represent with simple linear models. **Generative modeling:** autoencoders can be used for generative modeling by sampling from the latent space and decoding it to create new data samples. On the other hand, AEs have these major limitations that need to be handled by better modeling [30]. **Overfitting:** autoencoders can suffer from overfitting if the network capacity is too large or if the data are noisy. This factor can result in poor generalization of new data. **Computational cost:** training an autoencoder can be computationally expensive, particularly for large datasets or deep architectures. **Latent representation interpretation:** the latent representation learned by an autoencoder may not always be easily interpretable. This representation can make it difficult to understand the underlying patterns in the data. **Limited expressive power:** autoencoders may have limited expressive power compared to other NN architectures, such as convolutional neural networks or recurrent neural networks.

AEs are powerful DL techniques widely used for PQ event classification. Shi et al. [31] proposed an independent component analysis classification method for complex PQ disturbances using sparse autoencoder features. They achieved an average classification accuracy of 94.5% on the PQA database. Qiu et al. [32] proposed a modified S-transform and par-

allel stack sparse autoencoder method for PQ disturbance recognition. They achieved an overall accuracy of 97.6% on the simulation dataset and 95.9% on the measurement dataset. O'Donovan et al. [33] proposed a new deep-learning method for PQ disturbance classification using autoencoders. They achieved an accuracy of 96.5% on the PQA-S database. Liu et al. [34] proposed an automatic PQ disturbance diagnosis method based on a residual denoising convolutional autoencoder. It results in an overall accuracy of 97.5% on the simulation dataset and 98.7% on the measurement dataset. Liu et al. [35] proposed a compressed DL method for classifying multiple PQ events. They achieved an accuracy of 96.5% on the PQA-S dataset. Gonzalez-Abreu et al. [36] proposed a DL-based diagnosis method for PQ disturbances. They achieved an overall accuracy of 98.4% on the simulation dataset and 97.6% on the measurement dataset. Rodriguez et al. [37] proposed a PQ disturbance classification method using deep convolutional autoencoders and stacked LSTM recurrent neural networks. They achieved an overall accuracy of 95.4% on the PQA-S dataset. Chawda et al. [38] provided a comprehensive review of the PQ analysis using an autoencoder for its detection and identification. The above papers demonstrate that the autoencoder is a powerful technique for classifying PQ events. The overall classification accuracy achieved in these studies ranges from 94.5% to 98.4%. However, the accuracy may vary depending on the dataset, preprocessing techniques, and the autoencoder architecture.

AEs have shown great potential in various applications, including image and speech recognition, anomaly detection, and power quality (PQ) event classification. AEs can learn from the compressed representation of the input datasets and reconstruct it with high accuracy, which makes them a suitable tool for analyzing complex PQ signals. With advancements in ML and signal processing techniques, the AE-based classification of PQ events is expected to become more accurate and efficient. Some of the possible future developments in this area include improved architecture. Researchers are continuously exploring new AE architectures to enhance the performance of PQ event classification. Deep convolutional autoencoders (DCAEs) have shown promising results in identifying and classifying PQ events. Integration with other models is another aspect. AEs can be combined with machine learning models, such as support vector machines (SVMs) and neural networks, to improve classification accuracy. This integration can help to overcome some of the limitations of AEs, such as their tendency to overfit or underfit the training datasets.

Transfer learning is another well-known technique allowing AEs to learn from one dataset and apply the learned features to another. This technique can be useful in PQ event classification, where limited labeled datasets may be available for training. Real-time analysis of PQ events is essential for quickly identifying and mitigating PQ issues. AEs can be optimized for real-time analysis using low-latency architectures, such as lightweight convolutional autoencoders (LCAEs). Overall, the future of AE-based classification of PQ events looks promising. Ongoing research in this area likely leads to new and improved methods for analyzing PQ signals.

3.4. Generative Adversarial Network (GAN)

The GAN is a type of DL architecture, as shown in Figure 5, composed of two neural networks competing against each other in a zero-sum game framework [39]. The two networks are typically referred to as the generator and the discriminator. The generator aims to produce synthetic data indistinguishable from real data for the discriminator, while the discriminator's goal is to correctly classify each instance of data it sees as real or synthetic. The two networks are trained simultaneously in an adversarial manner, where the generator produces synthetic data, and the discriminator evaluates it. The generator adjusts its parameters based on the discriminator's feedback. This process continues until the generator produces synthetic data indistinguishable from real datasets for the discriminator. GANs have been used extensively for various tasks, including image generation, video synthesis, data augmentation, and even solving inverse problems such

as super-resolution and denoising. They have shown remarkable results applied to highly realistic synthetic datasets.

Real Image Sample

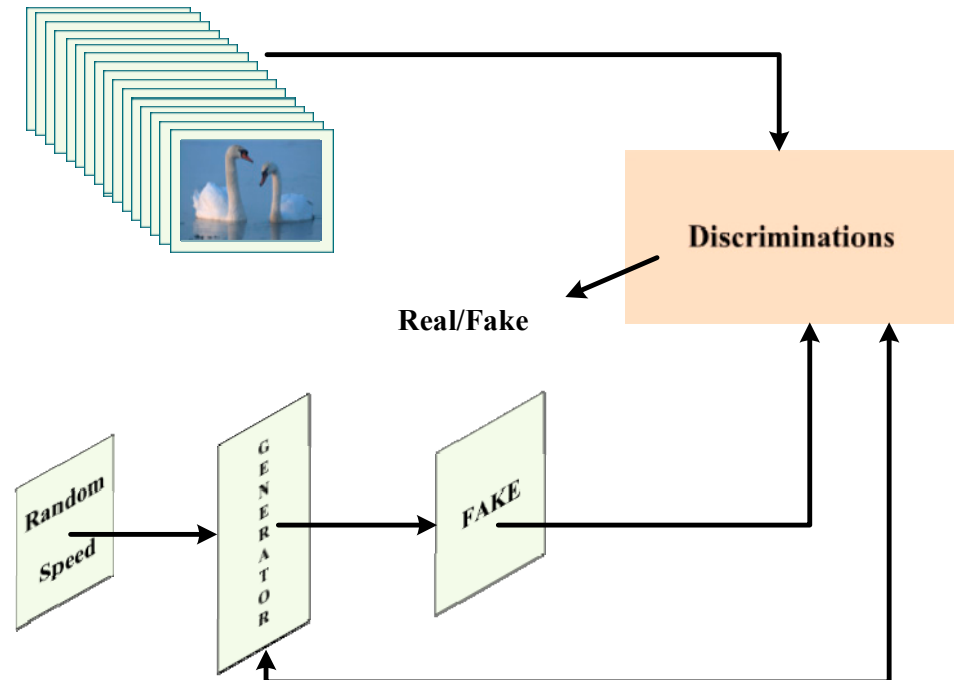


Figure 5. Architecture of generative adversarial network.

The GAN algorithm consists of two main components: the generator and the discriminator [40]. The various steps are as follows. Step-1 generator: the generator is an NN that inputs a random noise vector and produces a new data sample similar to the training datasets. The output of the generator is then passed to the discriminator. Step-2 discriminator: the discriminator is also an NN that takes a data sample as input and predicts whether the sample is real or generated by the generator. The output of the discriminator is a probability value between 0 and 1. Step-3 training: during training, the generator and discriminator are trained together using a two-player minimax game. The generator produces data samples so the discriminator can distinguish between real and fake data samples. Step-4 loss function: the loss function used to train the generator and discriminator is the binary cross-entropy loss. The generator tries to maximize this loss while the discriminator tries to minimize it. Step-5 updating: the weights of the generator and discriminator are updated after each training step using the backpropagation algorithm. This process continues until the generator can produce data samples indistinguishable from real data samples, or until a stopping criterion is met. The GAN algorithm is a powerful tool for generating new datasets and is applicable for productive tasks.

The major benefits of this method are as follows. GANs can generate high-quality, realistic samples that are difficult to distinguish from real data. They can capture the complexity and diversity of the input data, making them suitable for unsupervised learning tasks. GANs can be trained on large datasets, allowing them to learn complex distributions and patterns. They can be used for various applications, such as generating synthetic images, improving image quality, and creating realistic virtual environments. GANs are highly customizable and can be adapted to different domains and tasks by changing the generator and discriminator networks. However, the major limitations are as follows [41]. Training GANs can be challenging and requires a lot of computational resources. GANs can be unstable during training, leading to mode collapse, where the generator produces limited, low-quality data samples. The discriminator network can sometimes become too powerful,

leading to a situation where the generator cannot make meaningful progress. GANs may be difficult to interpret, as the generator and discriminator networks are complex and nonlinear. The generated samples can sometimes contain unwanted artifacts or patterns, making them unsuitable for certain applications. In conclusion, GANs are powerful and versatile tools for productive tasks, but have limitations and challenges. Careful design and training of the networks are crucial to achieving good results with GANs.

GANs have emerged as a powerful technique in PQ event classification due to their ability to learn complex patterns and generate new data samples. Cheng et al. [42] proposed an online power system event detection method using bidirectional GANs. They used the GAN architecture to learn the probability distribution of the power system data and generate samples of power system events. The proposed method achieved an overall accuracy of 97.3% for detecting three different types of events, i.e., fault, generator trip, and load shedding. In a recent paper, de Oliveira and Bollen [43] presented a comprehensive review of the application of DL methods in PQ analysis. They discussed using GANs for PQ event classification, and highlighted the importance of choosing GAN architecture and hyperparameters in achieving high classification accuracy. Jian and Wang [44] proposed a novel semisupervised PQ disturbance classification method using GANs. They used the GAN architecture to generate synthetic data samples and real data to train a classifier. The proposed method achieved an overall accuracy of 98.6% for classifying five different types of PQ disturbances. Cui et al. [45] proposed a novel technique for detecting and classifying multiple PQ disturbances using the Stockwell transform and DL. They used a GAN-based denoising autoencoder to preprocess the data, and a CNN to classify the PQ events. The proposed method achieved an overall accuracy of 98.3% for detecting six different types of PQ disturbances. Mohammadi et al. [46] proposed a protection scheme for high-impedance fault detection in distribution networks using a conditional GAN and convolutional classifier. The proposed technique achieved an overall accuracy of 98.5% for detecting high-impedance faults. Ma et al. [47] used a DL approach for PQ disturbance classification and compared the performance of different DL models, including GANs. The proposed GAN-based method achieved an overall accuracy of 96.2% for classifying eight different types of PQ disturbances. Overall, GAN-based methods have shown promising results in PQ event classification, achieving high accuracy in various studies. However, choosing GAN architecture and hyperparameters is crucial to the method's performance. Therefore, further research is needed to explore the optimal design of GAN-based models for PQ analysis.

GANs have shown great potential in various fields, including computer vision, PQ analysis, natural language processing, and signal processing. There has been growing interest in using GANs for PQ events classification in recent years, and the future scope for this application is promising. GANs can generate synthetic data to train ML models, improving the accuracy of classification models for PQ events. GANs can also be used to augment existing datasets, thereby increasing the amount of data available for training. In the future, GANs could be used to accurately classify more complex PQ events. They could also detect and classify PQ events in real-time, which is particularly useful in applications such as smart grids.

Furthermore, GANs could be used to develop automated PQ monitoring and control systems. By continuously analyzing and classifying PQ events, these systems could help to prevent power outages and equipment damage, thereby improving the reliability and efficiency of power systems. Overall, the future scope for GANs in PQ event classification is vast, and their potential applications could significantly improve the reliability and efficiency of power systems.

3.5. Deep Neural Network (DNN)

The DNN is an artificial NN, as illustrated in Figure 6, consisting of multiple layers of interconnected nodes, also known as artificial neurons [48]. The “depth” of a DNN refers to its number of layers, with each layer learning and passing on a more abstract representation

of the data to the next layer. Multiple layers allow a DNN to learn and model complex, nonlinear relationships between inputs and outputs, making them particularly useful for applications such as PQ analysis. DNNs use a combination of mathematical operations and weights to make predictions. The weights are learned through training, where the network is exposed to a set of inputs and the corresponding outputs, and the weights are updated to minimize the error between the network's predictions and the actual outputs. DNNs have revolutionized many fields, leading to substantial advancements in computer vision and natural language processing. They are also widely used in various industrial applications such as finance, healthcare, and autonomous systems.

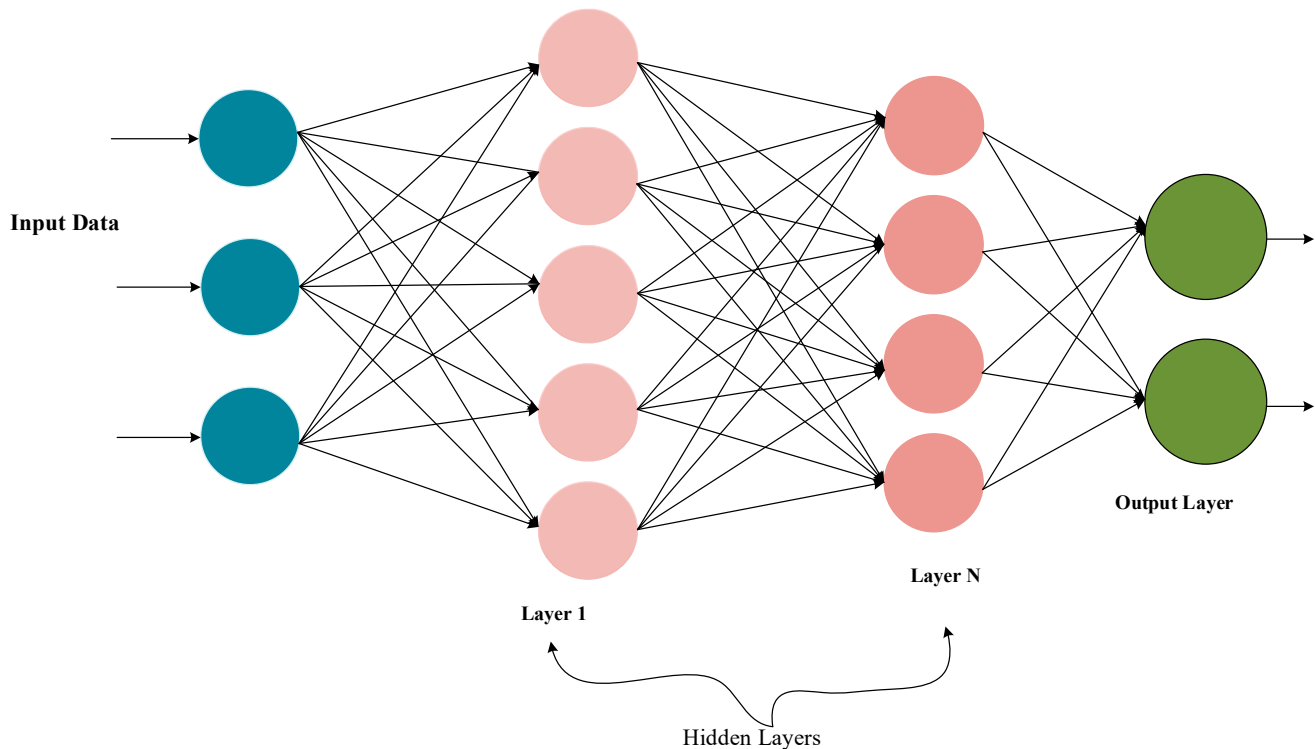


Figure 6. Architecture of deep neural network.

The algorithm for training a DNN involves the following steps [48]. Step-1 initialization: the network weights are initialized randomly. Step-2 forward propagation: the input datasets are passed through the network, layer-by-layer, using a set of activation functions to produce an output. Step-3 loss calculation: the difference between the network's output and the target output is calculated, producing a loss value. Step-4 backpropagation: the gradients of the loss concerning the network weights are calculated and used to update the weights in a direction that minimizes the loss. Step-5 repeat: steps 2–4 are repeated many times until the network converges, i.e., the loss reaches a minimum, and further weight updates no longer produce significant improvements.

The primary advantages of DNNs are as follows. Learning: it can model complex, nonlinear relationships between inputs and outputs. In addition, its ability to automatically learn useful features from raw datasets reduces the need for manual feature engineering. Scalability: DNNs can be easily extended to accommodate more layers and nodes, allowing them to learn more complex dataset representations. On the contrary, the primary disadvantages of DNNs are as follows [49]. Computational cost: training a DNN can be computationally expensive and requires large data and computational resources. Overfitting: DNNs can easily overfit to training data, reducing their ability to generalize to new, unseen datasets. Lack of interpretability: it can be difficult to understand how a DNN is making its predictions, making it challenging to diagnose and fix issues with the network.

Vulnerability to adversarial examples: DNNs can be easily fooled by small, carefully crafted inputs, known as adversarial examples, leading to incorrect predictions.

DL techniques have shown promising results in classifying PQ events. Balouji and Salor [50] proposed a novel approach for classifying PQ events using DL on event images. They used wavelet packet decomposition to transform the voltage and current signals into an image-like format. Then, a deep CNN was trained on these images to classify them. The proposed method achieved an overall accuracy of 97.8% on the IEEE PES PQ dataset. Mohan et al. [51] introduced Deep Power, an architecture for PQ disturbance classification. They combined one-dimensional CNN and long short-term memory (LSTM) networks to classify various PQ events. The proposed method achieved an overall accuracy of 97.3% on the same IEEE PES PQ dataset.

Liu et al. [52] comprehensively reviewed signal processing and DL techniques for monitoring and classifying power quality events. They discussed the advantages and disadvantages of different signal processing techniques, such as Fourier transform, wavelet transform, S-transform, and various deep-learning models such as CNN, LSTM, and autoencoder. They also highlighted the need for more research combining signal processing and DL techniques to classify PQ events. Liu et al. [53] proposed a compressed DL approach for classifying multiple PQ events. They used a deep NN with a compressed sensing layer to classify five PQ events. The proposed method achieved an overall accuracy of 98.3% on the same IEEE PES PQ dataset. Nandi et al. [54] presented a hyperbolic window S-transform-aided deep model-based PQ monitoring framework. They used a hyperbolic window S-transform to extract features from the voltage and current signals, and a deep NN for classification. The proposed technique achieved an overall accuracy of 99.5% on the same IEEE PES PQ dataset. The DL techniques have great potential for classifying PQ events. The proposed methods achieved high accuracy rates ranging from 97.3% to 99.5%. However, there is still a need for further research on the combination of signal processing and DL techniques to achieve better results.

DNNs have revealed promising results in many applications, including image and speech recognition, natural language processing, and, more recently, PQ event classification. These networks can effectively learn complex patterns and relationships in large datasets, making them ideal for handling large amounts of data generated by power systems. Here are some potential areas where DNNs could be applied. Enhanced classification accuracy: DNNs can be trained to classify PQ events more accurately than traditional classification methods, improving power system performance. Fault detection: DNNs can detect and classify faults in power systems, which can help improve the reliability of power distribution and transmission systems. Feature extraction: DNNs can automatically extract relevant features from raw data, improving classification algorithm accuracy and reducing manual data preprocessing. Real-time monitoring: DNNs can be trained to operate in real-time, allowing them to monitor PQ events as they occur and quickly respond to any issues that arise. Predictive maintenance: DNNs can be used to predict potential failures in power systems, allowing for proactive maintenance to prevent costly downtime. Overall, the future of DNNs for the classification of PQ events looks promising, with potential benefits ranging from improved reliability to reduced costs and increased efficiency in power systems.

3.6. Deep Belief Network (DBN)

The DBN is a generative artificial NN used for unsupervised learning [55]. It comprises multiple layers of restricted Boltzmann machines (RBMs), as shown in Figure 7, shallow and two-layer networks that can learn to extract features from the input datasets. The outputs from one RBM are used as inputs to the next, creating a deep network. DBNs are trained using a greedy layer-wise unsupervised learning algorithm, where each layer is trained to reconstruct the inputs fed to it, one layer at a time, from the bottom up. The algorithm allows the DBN to learn hierarchical representations of the data, with each layer capturing more complex features than the previous one. Once the DBN is trained, it can be

used for various tasks, such as dimensionality reduction, feature extraction, and generative modeling. Additionally, the learned features can be used as inputs to a supervised learning algorithm for classification or regression tasks. In addition to PQ analysis, DBNs have been used for various applications, including image and speech recognition, recommendation systems, and natural language processing.

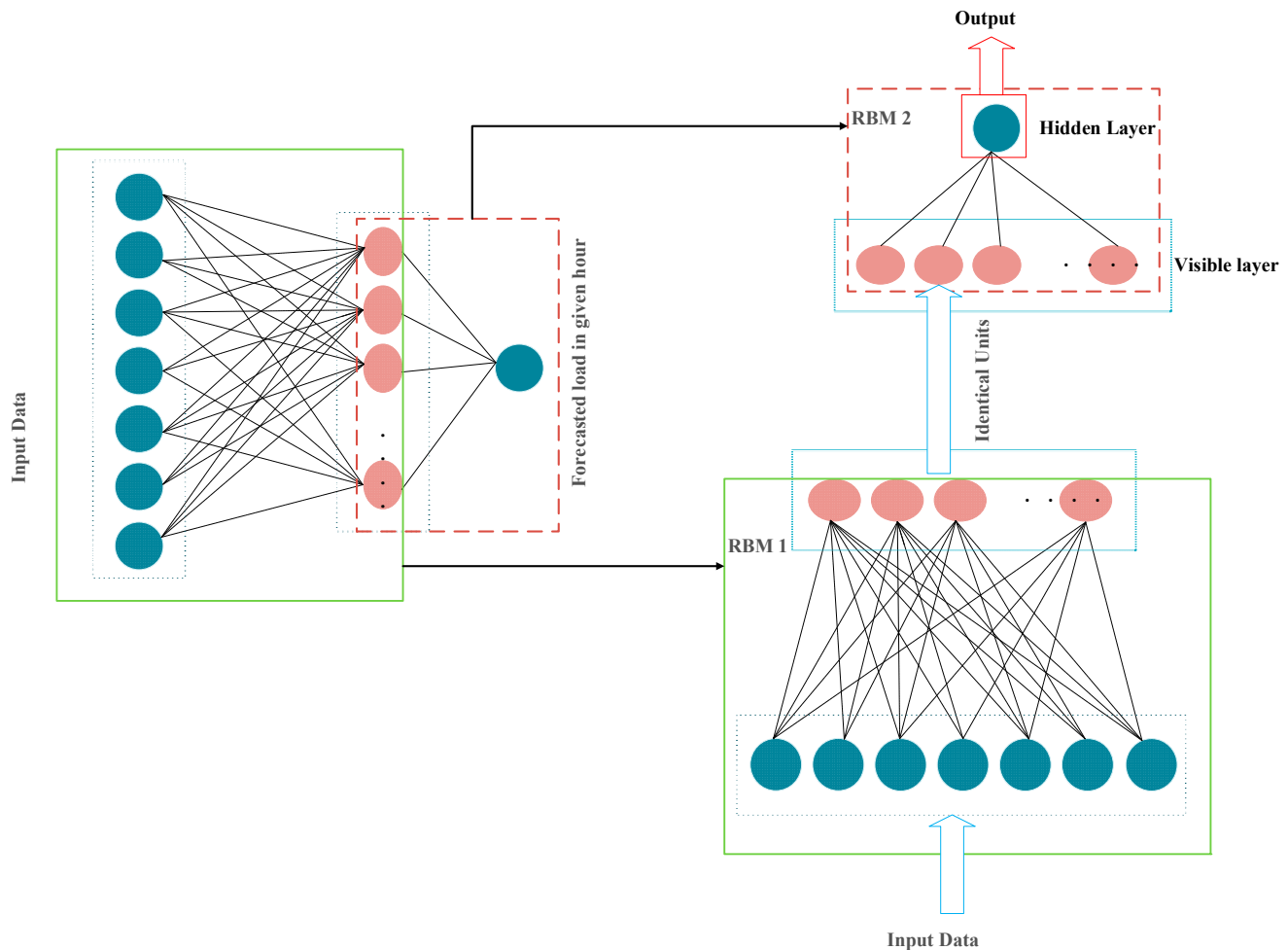


Figure 7. Architecture of deep belief network.

The algorithm for training a DBN consists of the following steps [55]. Step-1 pretraining: each layer of the DBN is trained as an RBM in a greedy layer-wise manner, starting with the first layer and proceeding to the next layer until all layers have been trained. During pretraining, the RBMs are trained using the contrastive divergence algorithm to learn the weights that capture the input–output dependencies. Step-2 fine-tuning: after pretraining, the DBN is fine-tuned using a supervised learning algorithm, such as backpropagation, to adjust the weights based on the target outputs.

Three major advantages of DBNs are as follows. Greedy layer-wise training: DBNs are trained layer-by-layer, allowing each layer to learn a more complex representation of the data than the shallow networks. Unsupervised pretraining: DBNs can learn meaningful representations of the data without needing labeled data, making them well-suited for unsupervised learning tasks. Hierarchical representation: the learned representations are hierarchical, meaning each layer captures more complex features than the previous layer. The four major disadvantages are as follows [56]. Computational complexity: training a DBN can be computationally expensive, especially for large datasets and deep networks. Overfitting: DBNs can overfit the training data, especially if the network is deep and the training data are limited. Initialization sensitivity: the pretraining step is sensitive to the

initialization of the weights and can lead to poor results if not initialized correctly. Limited ability to capture nonlinear relationships: DBNs are limited in their ability to capture nonlinear relationships in the data compared to other deep-learning architectures, such as CNNs.

Deep belief networks (DBNs) have shown promising results in classifying power quality (PQ) events. This review discusses several recent studies that use DBNs for PQ event classification. Mei et al. [57] proposed an online recognition method for voltage sags using a DBN. The proposed method achieved an accuracy of 98.53% on the dataset of voltage sags. Sekar et al. [58] proposed an improved PQ disturbance detection method using a DBN. The proposed method achieved an accuracy of 98.87% on the dataset of PQ disturbances.

Soumya et al. proposed an islanding PQ detection method using a lighting search optimization with a DBN model on distributed generation systems. The proposed method achieved an overall accuracy of 96.2% on the dataset of islanding PQ events. Deng et al. [59] proposed a sequence-to-sequence DBN architecture based on bidirectional gated recurrent units for type recognition and time location of combined PQ disturbances. The proposed method achieved an overall accuracy of 99.3% on the dataset of PQ disturbances. Cai et al. [60] proposed a PQ disturbance classification method using the Wigner–Ville distribution and deep CNNs. The proposed method achieved an overall accuracy of 98.2% on the dataset of PQ disturbances. Wang et al. [61] proposed a novel DBN-based fault diagnosis approach for a chemical process. The proposed method achieved an overall accuracy of 99.2% on the dataset of fault diagnosis. Overall, the above studies demonstrate the effectiveness of DBNs in PQ event classification, achieving high classification accuracies on different PQ event datasets. However, the choice of network architecture and optimization techniques can influence the performance of the DBN. DBNs are promising for PQ event classification, and further studies are needed to investigate the generalization ability of DBNs on different PQ event datasets.

DBNs have shown promising results in various applications, including PQ event classification. DBNs are a deep-learning model that utilizes multiple layers of hidden units to extract hierarchical features from the input datasets. In the context of PQ event classification, DBNs can automatically learn and extract relevant features from the raw power signal. This property can improve classification accuracy compared to traditional methods that rely on hand-engineered features. The future scope of DBNs for power quality event classification is promising.

However, potential areas where DBNs can be further explored are as follows. Improved accuracy: researchers can explore different architectures and training methods to improve the accuracy of DBNs for PQ event classification. Large-scale deployment: DBNs can be trained on large-scale datasets and deployed in real-world power systems to provide accurate and reliable power quality event classification. Multiclass classification: DBNs can be extended to classify multiple PQ events simultaneously. The mentioned characteristic can improve the efficiency and effectiveness of PQ monitoring systems. Online learning: DBNs can be trained online, which allows the model to adapt to changing power system conditions in real-time. This learning can improve the accuracy of the classification and reduce false alarms. Interpretability: DBNs are considered to be black-box models. Researchers can explore different methods to make the model more interpretable, which can help power system operators to understand the reasons behind the detected power quality events. DBNs offer a promising avenue for PQ event classification, and further research in this area can lead to more accurate and efficient PQ monitoring systems.

3.7. Long Short-Term Memory (LSTM)

LSTM is an RNN type, as shown in Figure 8, designed to handle the problem of vanishing gradients in traditional RNNs [62]. It is a kind of RNN that is particularly well-suited for sequential data and tasks requiring long-term memory, such as speech recognition and natural language processing. LSTMs were introduced in the 1990s and

have become one of the most widely applied DL architectures for sequence-to-sequence problems. They work by introducing memory cells and gates that control the flow of information into and out of the cell state, allowing the model to selectively preserve or forget information over long sequences. LSTMs are trained through backpropagation and gradient descent, just like other NNs. The model adjusts its weights and biases during training to minimize the prediction error. Once trained, the LSTM can generate predictions for new, unseen data. LSTMs are widely applied in various applications such as speech recognition, machine translation, and sentiment analysis. They have also been used in various domains, including finance, healthcare, and retail, to analyze sequential data and extract insights that can be used to make informed decisions.

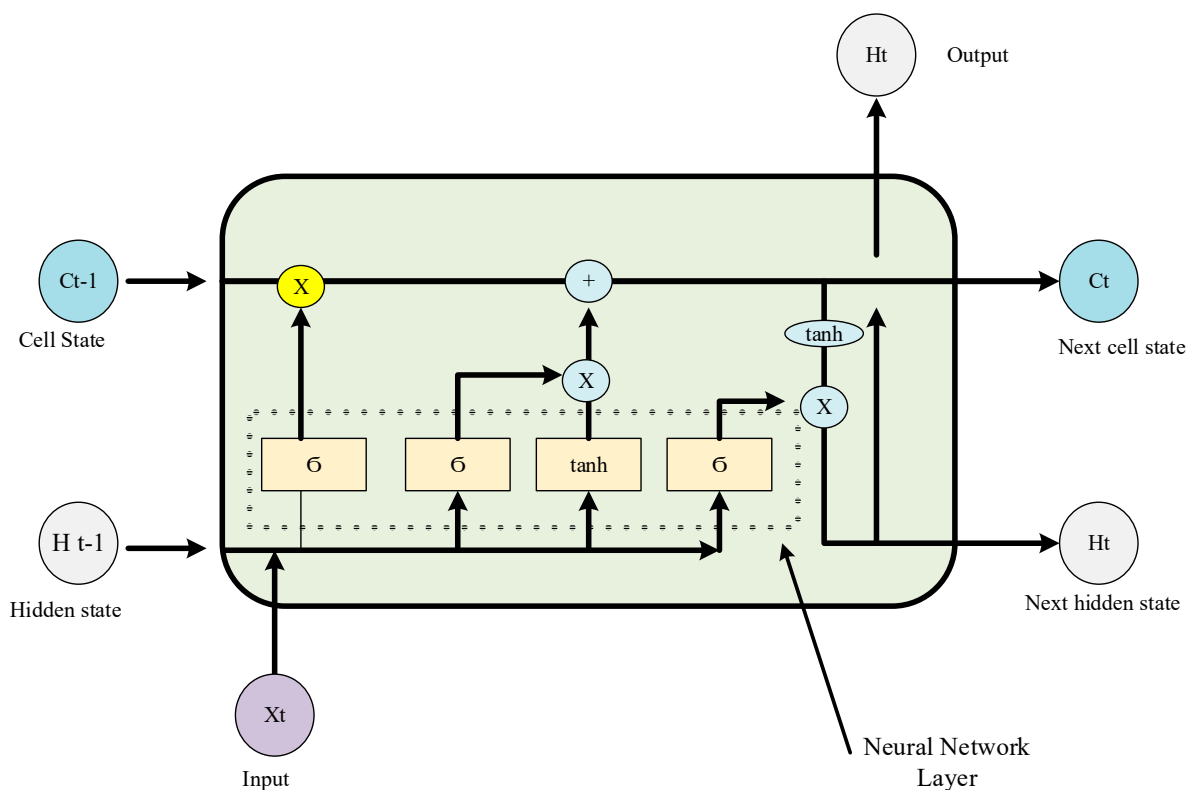


Figure 8. Architecture of long short-term memory.

The LSTM algorithm is an RNN designed to handle the problem of vanishing gradients in traditional RNNs [62]. The basic idea behind the LSTM algorithm is to use memory cells and gates to control the flow of information in and out of the cell state. The algorithm of LSTM can be summarized as follows. Step-1: initialize the cell state with an initial value and the hidden state with random weights. Step-2: process each time step of the input sequence by updating the cell state and hidden state based on the previous time step's cell state, hidden state, and the current time step's input. Step-3: the updated cell and hidden states predict the current time step. Step-4: repeat the process for each time step in the input sequence.

The three major advantages of the LSTM algorithm are as follows. Handling of vanishing gradients: LSTMs are designed to handle the problem of vanishing gradients in traditional RNNs, allowing the model to preserve information over long sequences. Ability to model long-term dependencies: LSTMs are particularly well-suited for tasks that require the model to remember information from earlier time steps, making them well-suited for sequence-to-sequence problems. Robustness to noisy data: LSTMs are robust to noisy or missing data, making them suitable for real-world applications where data are often noisy or incomplete. The three major disadvantages of the LSTM algorithm are as follows [63]. Computational complexity: LSTMs can be computationally expensive

due to their large number of parameters, making them difficult to train on large datasets. Lack of interpretability: LSTMs are often considered “black boxes” due to their complex structure, making it difficult to understand why the model makes certain predictions. Overfitting: LSTMs can be prone to overfitting if not properly regularized, leading to poor generalization performance on new, unseen data. The LSTM algorithm is a powerful tool for processing sequential data and modeling long-term dependencies. However, it can be computationally complex and prone to overfitting if not used carefully.

LSTM is an RNN type with promising results in various applications, including PQ event classification. Junior et al. [64] proposed a PQ disturbance classification method using a CNN and LSTM network. The proposed method achieved an overall accuracy of 97.34% on the benchmark dataset. Abdel Salam et al. [65] developed a PQ event categorization method using LSTM networks. They utilized a new feature extraction technique based on the wavelet packet transform (WPT). Their method achieved an overall accuracy of 98.8% on the benchmark dataset. Rajiv and Tripathi [66] proposed a deep hybrid learning approach based on LSTM-CNN for power quality event classification. Their method achieved an overall accuracy of 99.3% on the benchmark dataset. Özer et al. [67] proposed a CNN/Bi-LSTM-based DL algorithm using spectrogram images for PQ event classification. Their method achieved an overall accuracy of 97.2% on the benchmark dataset. Rodriguez et al. [68] proposed a PQ event classification method using the Hilbert–Huang transform and LSTM networks. Their method achieved an overall accuracy of 95.4% on the benchmark dataset. Garcia et al. [69] compared three DL methods for detecting PQ disturbance and classification: CNN, LSTM, and CNN-LSTM. Their results showed that the CNN-LSTM method achieved the highest accuracy of 98.9% on the benchmark dataset. Manikonda et al. [70] proposed a PQ event classification method using LSTM networks. Their method achieved an overall accuracy of 97.7% on the benchmark dataset. Skydt et al. [71] proposed a probabilistic sequence classification approach based on LSTM networks for early fault prediction in distribution grids. Their method achieved an overall accuracy of 92.3%. Balouji et al. [72] proposed an LSTM-based deep-learning method for voltage dip classification. Their method achieved an overall accuracy of 98.5% on the benchmark dataset. Chiam et al. [73] proposed a global attention-based LSTM for noisy PQ disturbance classification. Their method achieved an overall accuracy of 98.9% on the benchmark dataset. The reported classification accuracies ranged from 92.3% to 99.3%, indicating that LSTM-based methods can achieve high accuracy in the PQ analysis.

LSTM is an RNN type useful for modeling data sequences over time. PQ events often occur over time and can be represented as time-series data. LSTM has shown promising results in the classification of PQ events. By training an LSTM model on labeled PQ data, the model can learn to classify different PQ events based on the patterns in the time-series data. In the future, LSTM and other DL techniques play an increasingly important role in classifying PQ events. As the amount of data generated by smart grids and other sources continues to grow, DL algorithms can handle this data’s increasing complexity and volume. Moreover, integrating LSTM with other ML techniques for clustering and feature extraction can improve classification results. Additionally, advancements in hardware and software technologies enable faster and more efficient training of DL models, further enhancing the performance of LSTM for classifying PQ events.

3.8. Deep Autoencoder (DAE)

An autoencoder (AE) is a type of NN, as shown in Figure 9, comprising two components, an encoder, and a decoder, which adopts unsupervised learning [74]. The encoder compresses the input data into a lower-dimensional representation (the encoding), while the decoder tries to reconstruct the original input data from this encoding. A DAE is a variant of the traditional autoencoder that uses multiple layers in both the encoder and the decoder. Due to this multilayer topology, it can learn more complex representations of the input data using multiple layers. The above can lead to better reconstructions and improved performance on tasks such as dimensionality reduction, anomaly detection, and

generative modeling. In general, the architecture of a DAE can be seen as a stack of multiple shallow autoencoders, where each shallow autoencoder learns a layer of representation. The encoding of each layer is used as the input to the next layer, and the decoder works in reverse, reconstructing the input from the top most encoding layer to the original data.

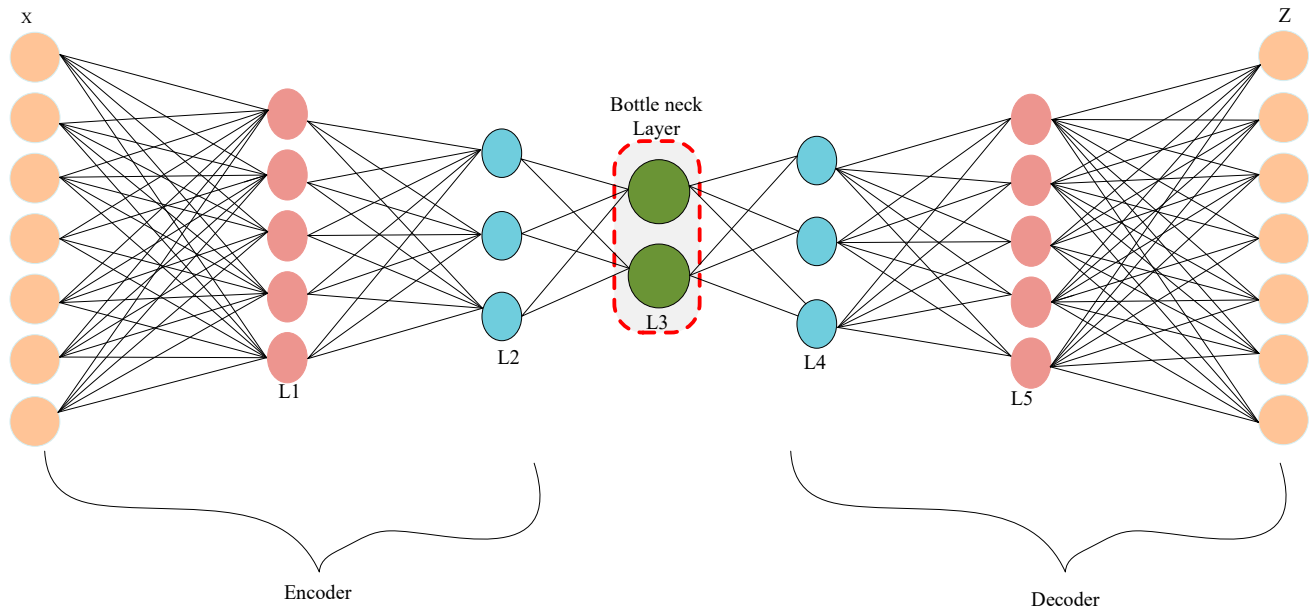


Figure 9. Architecture of deep autoencoder.

A DAE is a type of NN that is used for unsupervised learning. The main goal of an AE is to learn a compact representation (encoding) of the input data and then use this encoding to reconstruct the original data. The algorithm of a DAE can be summarized as follows [74]. Step-1: the input data are passed through an encoder network that consists of multiple dense layers. The output of the encoder network is the encoding of the input data. Step-2: the encoding is then passed through a decoder network that consists of multiple dense layers. The output of the decoder network is the reconstructed version of the original data. Step-3: the loss between the original and reconstructed data is calculated using a reconstruction loss function such as mean squared error (MSE) or binary cross-entropy (BCE). Step-4: the weights of the encoder and decoder networks are updated using an optimization algorithm, such as stochastic gradient descent (SGD) or Adam, to minimize the reconstruction loss. Step-5: the process is repeated until the reconstruction loss converges to an acceptable level.

The advantages of the DAE are as follows. Deep autoencoders can learn compact and meaningful representations of the input data. AEs can be used for dimensionality reduction, data denoising, and anomaly detection tasks. They can handle nonlinear relationships between the input and output data. AEs can be easily combined with other DL models to form hybrid models. The primary disadvantages of deep autoencoders are as follows [75]. DAEs are computationally expensive compared to other dimensionality reduction techniques, such as PCA. Autoencoders may struggle to preserve the global structure of the input data in case the encoding and decoding networks are too small. AEs can be sensitive to the choice of the reconstruction loss function and the number of layers in the encoder and decoder networks. Additionally, it may not perform well on high-dimensional and complex data, as they require lots of data to train effectively.

DL methods have recently gained significant attention in PQ event classification due to their high accuracy and efficiency. Sahani and Dash (2021) proposed a deep convolutional stack autoencoder of process-adaptive VMD data with a robust multikernel RVFLN for PQ event recognition [76]. They achieved a classification accuracy of 99.11% using the proposed method. Ge et al. (2020) presented a deep feature clustering approach for seeking

patterns in daily harmonic variations [77]. The proposed method achieved an accuracy of 98.3% for classifying harmonic patterns.

Liu et al. (2019) used a compressed DL approach to classify multiple PQ events [78]. They achieved an overall accuracy of 97.35% using the proposed method. Dash et al. (2022) proposed an adaptive morphological filter with a deep-learning algorithm for multiple PQ disturbances analysis in a photovoltaic integrated direct current microgrid [79]. The proposed method achieved an overall accuracy of 99.5%. Ge et al. (2021) proposed an unsupervised DL and analysis approach for harmonic variation patterns using big data from multiple locations [80]. The proposed method achieved an overall accuracy of 97.2% for identifying harmonic patterns. De Oliveira et al. (2021) proposed a DL method with manual postprocessing for identifying spectral patterns of waveform distortion in PV installations [81]. They achieved an overall accuracy of 97.8% using the proposed method. Salles et al. (2022) proposed an analytics approach for waveform distortion variations in railway pantograph measurements by DL [82]. They achieved an overall accuracy of 96.5% using the proposed method. Overall, these research papers demonstrate the effectiveness of DL methods for PQ event classification, with overall accuracy ranging from 96.5% to 99.5%. The proposed methods can identify various types of PQ events and can be applied to real-world applications. However, there is still room for improvement in the robustness and generalization of the proposed models.

DAEs have shown promising results in various applications of image processing, natural language processing, and anomaly detection. They have recently been used for event classification in PQ monitoring systems. PQ events are disturbances in the electrical power supply that can cause equipment malfunction, loss of productivity, and in extreme cases, life-threatening situations. PQ monitoring systems detect and classify these events to prevent equipment damage and ensure a safe and reliable power supply. DAEs can be used for PQ event classification by training on labeled data of different event types. The AE learns to encode the input signal into a compressed representation, and then decodes it back into the original signal. The compressed representation can then be used to classify the event type. The future scope of using DAEs for PQ event classification is promising. A few potential developments exist. Incorporating more features: DAEs can be trained on waveform features and other electrical parameters such as voltage and current harmonics, power factor, and frequency to improve event classification accuracy. Real-time detection: real-time detection of PQ events is crucial for preventing equipment damage and ensuring PQ reliability. DAEs can be extended to real-time event detection by incorporating online learning techniques. Transfer learning: transfer learning can apply pretrained AE models to new datasets with different PQ event types. The above fact can reduce the need for extensive labeled data, and improve the efficiency of the event classification process. Integration with other machine learning techniques: DAEs can be integrated with other ML techniques, such as supervised and unsupervised learning, to improve the accuracy of PQ event classification. In conclusion, the future scope of using DAEs for PQ event classification is vast and promising. Continued research and development can improve PQ monitoring systems and a safe and reliable power supply.

3.9. Self-Organizing Map (SOM)

The SOM is a type of ANN, as illustrated in Figure 10, and is based on unsupervised learning [83]. The main goal of an SOM is to reduce the high-dimensional data into a low-dimensional representation while preserving the topological structure of the data. In a SOM, data points are mapped to nodes in a grid-like structure, where similar data points are grouped and assigned to nearby nodes. This results in the formation of clusters, which can be interpreted as the reduced representation of the original data. The SOM algorithm updates the weights of the nodes in such a way that the similarity between a data point and the weight vector of a node is maximized. SOMs are widely used in various fields, such as data visualization, compression, pattern recognition, and clustering. They are especially useful when working with complex, high-dimensional data, as they can effectively capture

the underlying structure and relationships in the data. Overall, the SOM is a powerful tool for unsupervised learning and data analysis, and it continues to be an active area of research in ML.

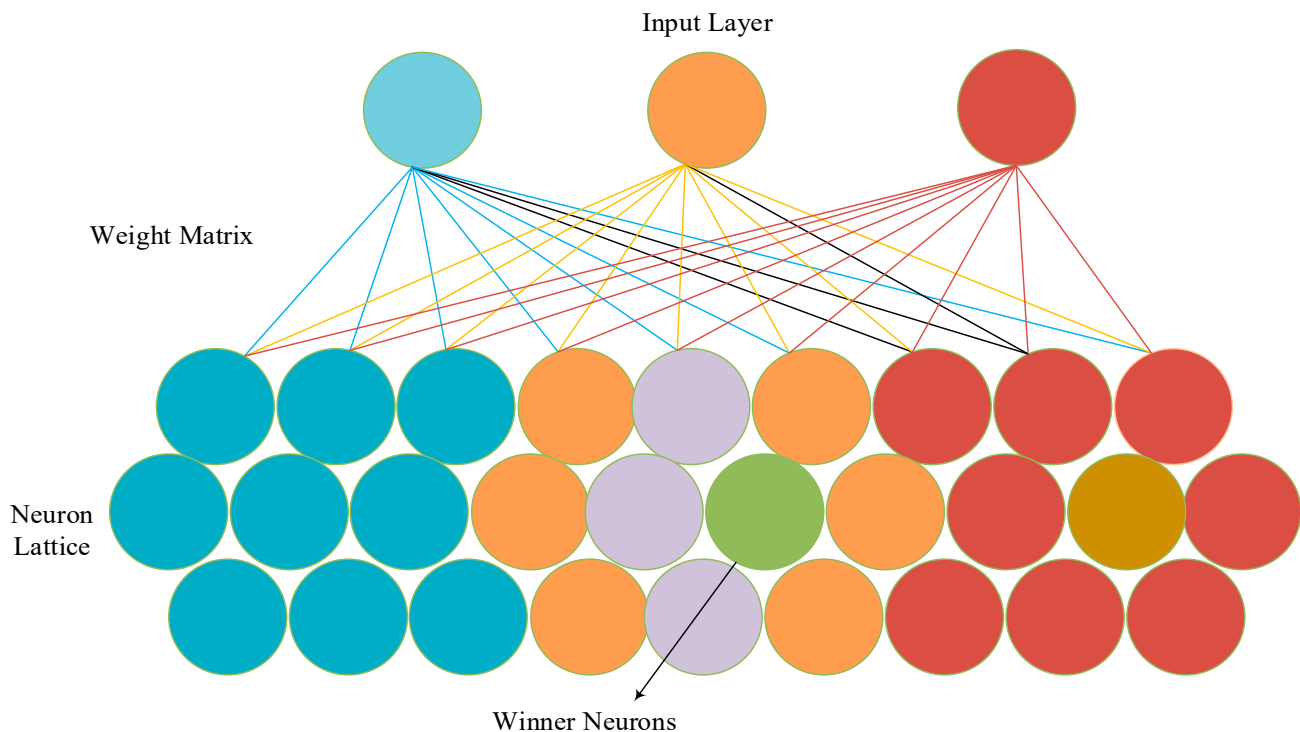


Figure 10. Architecture of self-organizing map.

The SOM algorithm is an unsupervised learning algorithm that can be broken down into steps [83]. Step-1 initialization: start by randomly initializing the weights of the nodes in the grid-like structure. Step-2 training: for each iteration, select a data point from the input dataset and calculate the Euclidean distance between the data point and the weight vectors of all nodes. The node with the closest weight vector is considered the best matching unit (BMU). Step-3 updating weights: the weights of the BMU and its surrounding nodes are updated to reduce the distance between the BMU and the data point. This process moves the weights toward the data point and decreases the update rate over time. Step-4 repeat: repeat the training process for a specified number of iterations or until the algorithm converges.

The four major advantages of SOM are as follows. Data visualization: the SOM effectively visualizes complex, high-dimensional data in a two-dimensional grid. This property makes it easier to understand the structure and relationships in the data. Clustering: the SOM can cluster data points into groups based on similarity. The nodes in the SOM can be interpreted as the centroids of the clusters. Robustness: the SOM is a robust algorithm that can handle noisy or missing data, and it can also handle data with different scales and distributions. Efficient: the SOM is an efficient algorithm, especially for small- to medium-sized datasets, as it can be trained in minutes or hours. The four major disadvantages of the SOM are as follows [84]. Limited interpretability: the SOM can be difficult, especially when working with complex data. Understanding the relationships between the nodes and the original data can be challenging. Lack of model transparency: the SOM is a black-box model, which means it is not always clear how the algorithm arrived at a particular result. Overfitting: the SOM can overfit the data, especially when working with small datasets, as it can capture the noise in the data instead of the underlying structure. Limited generalization: the SOM is not designed for generalization and can struggle to make predictions on new, unseen data. In conclusion, the SOM is a powerful tool for

unsupervised learning, but has limitations. It is important to understand the strengths and weaknesses of the algorithm to use it effectively.

The SOM has been widely used in classifying PQ events. Bentley et al. [85] proposed an SOM-based method for identifying the source of PQ disturbances. They achieved an overall classification accuracy of 95% on their dataset. Similarly, de la Rosa et al. [86] used higher-order cumulants and SOM for the amplitude–frequency classification of PQ transients, achieving an overall accuracy of 96.7%. Huang et al. [87] proposed a method based on S-transform and SOM neural networks for PQ disturbance recognition. Their proposed method achieved an overall accuracy of 97.2%. Kow et al. [88] reviewed the performance of AI and conventional methods in mitigating PV grid-tied related power quality events. They found that SOM-based methods have shown promising results in PQ events classification. Spassiani and Mason [89] applied SOM to classify the meteorological origin of wind gusts in Australia, achieving an overall accuracy of 85%. Martin-Fernandez et al. [90] proposed an SOM-based method for detecting emerging faults in power transformers. They achieved an overall accuracy of 91.7% on their dataset. Finally, De la Hoz et al. [91] proposed a feature selection method based on multiobjective optimization and hierarchical SOM for network anomaly detection. They achieved an overall accuracy of 98.1%. Overall, SOM has shown great potential for PQ events classification, achieving high accuracy rates in several research papers. The cited papers demonstrate the effectiveness of SOM-based methods for identifying the source of disturbances, classifying transients, detecting emerging faults, and recognizing PQ disturbances.

The SOM is an unsupervised ML technique for various data mining applications, including pattern recognition, clustering, and dimensionality reduction. In PQ event classification, SOM is useful for identifying and characterizing different types of PQ disturbances. The future scope of SOM for PQ events classification is promising, as there are several potential applications where SOM can be utilized to improve power system performance and reliability. A few possible future directions for SOM in PQ events analysis can be presented as follows. Improved classification accuracy: one of the main challenges in PQ event classification is achieving high accuracy, especially when dealing with complex and dynamic disturbances. The SOM can improve classification accuracy by leveraging its ability to represent high-dimensional data in a low-dimensional space, facilitating easier interpretation and visualization of the data. Real-time event detection: another potential application of SOM in PQ events classification is real-time event detection. Training SOM on a large dataset of PQ events can quickly identify and classify new events as they occur, allowing for faster response times and more efficient management of power systems. On-line learning: the SOM can be adapted for online learning, which means it can continuously update its internal representations based on new data as it becomes available. This property can be useful when the underlying characteristics of PQ events change over time, such as during system upgrades or changes in the load profile. Integration with other machine learning techniques: the SOM can be combined with other ML techniques, such as NNs and decision trees, to improve classification accuracy and facilitate a more comprehensive analysis of PQ events. This fact can lead to a better understanding of the underlying causes of PQ disturbances, and enable more effective mitigation strategies. In summary, the future scope of SOM for PQ events classification is vast, and we expect to see more innovative approaches that leverage the power of SOM to address the challenges of PQ event classification.

3.10. Restricted Boltzmann Machine (RBM)

The RBM is a generative stochastic ANN, as shown in Figure 11, which can learn a probability distribution over its set of inputs [92]. These are often used as building blocks for DL algorithms and can be used for various tasks, including dimensionality reduction, collaborative filtering, and feature learning. These are composed of visible units, which represent the input data, and hidden units, which capture the underlying structure of the data. During training, the RBM learns the relationship between the visible and hidden

units and can generate new data samples similar to the training data. The training process involves adjusting the weights and biases of the network to maximize the likelihood of the training data under the model. RBMs have received much attention in recent years because they can be trained using efficient algorithms, such as contrastive divergence, and they are relatively simple to understand and implement. They are also computationally efficient, making them well-suited for large-scale applications.

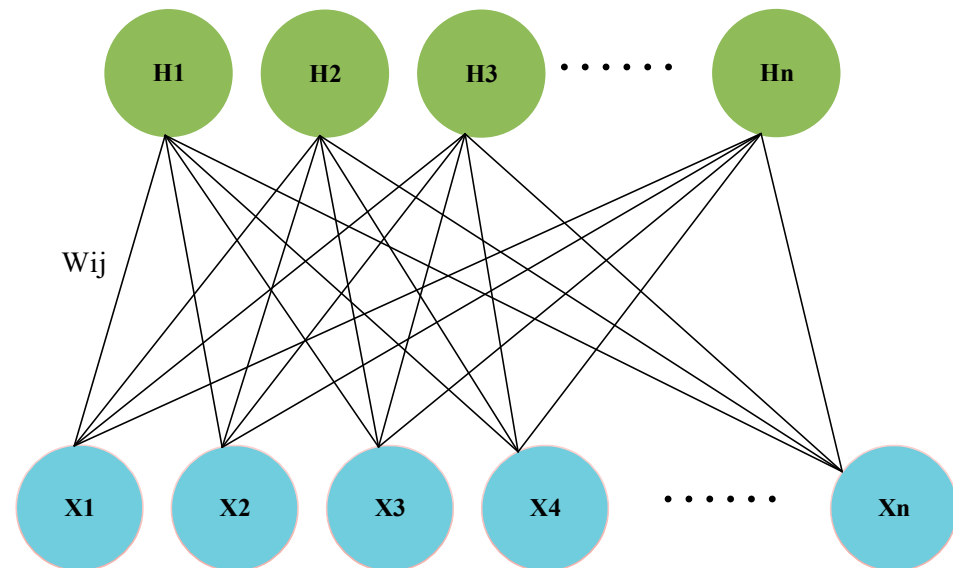


Figure 11. Architecture of restricted Boltzmann machine.

The training algorithm for an RBM is typically based on gradient-based optimization, such as stochastic gradient descent (SGD). The basic steps of the algorithm are as follows [92]. Step-1: initialize the weights and biases of the network randomly. Step-2: pass the input data through the network to obtain the activations of the hidden units. Step-3: use the activations of the hidden units to reconstruct the input data. Step-4: compute the reconstruction error between the original and reconstructed input. Step-5: use the reconstruction error to update the weights and biases of the network using gradient descent. Step-6: repeat steps 2–5 for multiple iterations until the reconstruction error converges or a stopping criterion is met. The training process of an RBM can be seen as learning the underlying probability distribution of the data, which is represented by the weights and biases of the network. Once the RBM is trained, it can be used for various tasks, such as generating new data samples, feature extraction, or dimensionality reduction.

The three major advantages of RBMs are as follows. Efficient and fast training: RBMs can be trained using efficient algorithms, such as contrastive divergence, which makes them well-suited for large-scale applications. Good performance on certain tasks: RBMs have performed well on certain tasks, such as collaborative filtering and dimensionality reduction. Simple and interpretable: RBMs are relatively simple to understand and implement, making them a good starting point for exploring DL algorithms. The three prominent disadvantages of RBMs are as follows [93]. Limited capacity: RBMs have a limited capacity to model complex data distributions, which makes them less suitable for certain tasks. No end-to-end training: RBMs are typically trained in a greedy layer-by-layer manner, which means they cannot be trained end-to-end like a deep neural network. Limited scalability: RBMs can be trained efficiently on large datasets, but do not scale well to large datasets or complex architectures.

RBM have been widely used to classify PQ events in recent years due to their ability to capture complex data features. Liu et al. [94] proposed a PQ event monitoring and classification method using DL techniques, including RBMs. They applied RBMs to extract features from PQ data and then used a support vector machine (SVM) for classification.

The proposed method achieved an overall classification accuracy of 95%. Khetarpal and Tripathi [95] used a semisupervised deep belief network (DBN) for PQ disturbances classification. They used a combination of RBMs and SVMs for classification and achieved an overall accuracy of 96%.

Fiore et al. [96] proposed an RBM-based network anomaly detection method, which could also be applied to PQ event classification. They achieved an overall accuracy of 99.3% for network anomaly detection, but the accuracy for PQ events classification was not reported. Gao et al. [97] proposed an adaptive wavelet threshold and DBN-ELM hybrid model for PQ disturbance classification under noisy conditions. They achieved an overall classification accuracy of 98%. De Oliveira and Bollen [98] reviewed DL techniques for PQ analysis and proposed an RBM-based approach for voltage dip classification. The proposed method achieved an overall accuracy of 99%. Soumya et al. [99] proposed an RBM-based lighting search optimization with a DL model for islanding PQ detection in distributed generation systems. The proposed method achieved an overall accuracy of 96.7%. Bayrak and Yilmaz proposed an AI-based PQ disturbance detection and classification method in smart grids using RBMs. They achieved an overall classification accuracy of 93.7%. Finally, Xi et al. [100] proposed a KF-ML-aided DBN method for identification of multiple types of PQ disturbances and time locations. The proposed method achieved an overall accuracy of 97.5% for type identification and 97.3% for time location. In summary, the reviewed papers show that RBMs have been effectively applied for power quality event classification, achieving high overall accuracy rates ranging from 93.7% to 99%. However, it is important to note that the performance of RBMs depends on the quality of the input data, the choice of hyperparameters, and the specific application.

RBM have been widely used for various ML tasks such as feature learning, unsupervised pretraining, collaborative filtering, and classification. RBMs have also shown promising results for event classification in PQ analysis. In the future, RBMs could continue to be used for PQ event classification with the following potential advancements. Improved accuracy: RBMs could be combined with other DL techniques, such as CNN or RNN, to improve the accuracy of PQ event classification further. Real-time event detection: with the increasing demand for real-time monitoring and control of power systems, RBMs could be used to develop algorithms for real-time detection of PQ events. The above purpose could be achieved by integrating RBMs with suitable hardware and software platforms. Interpretable models: the interpretability of ML models is a critical aspect of the power industry. RBMs can be designed to generate interpretable models that provide insights into the underlying characteristics of PQ events. Transfer learning: RBMs can be used for transfer learning, where the knowledge gained from one power system can be transferred to another, reducing the need for large amounts of labeled data. Explainable AI (XAI) is an emerging field that aims to make ML models more transparent and interpretable. RBMs can be used as part of XAI techniques to provide insights into the model's decision-making process. Overall, RBMs have the potential to be used for various applications in PQ analysis, and further research and development could lead to significant advancements in the field.

4. Critical Findings and Analysis

Correct PQ event classification ensures countermeasure action to impact the power system's efficient and reliable operation. DL has shown promising results in this field, but some limitations and challenges must be addressed. The three major limitations can be presented, confined to the DL application to PQ analysis, as follows.

- Data availability: one of the main limitations of DL for PQ event classification is the lack of sufficient labeled data. Collecting and labeling PQ event data can be challenging and expensive, hindering the development and performance of deep-learning models.
- Computational requirements: DL models are computationally intensive and require significant computational resources to train and run. This fact can be a challenge for power systems with limited computing capabilities.

- **Interpretability:** DL models are often considered “black boxes,” meaning it can be complex to interpret the results and realize how the model arrived at its decision. That is why it is challenging to identify the root cause of PQ events and take appropriate corrective action.

Three major recommendations are suggested to handle the DL model’s specific limitations, particularly for general classification applications.

- **Data augmentation:** these techniques can be applied to generate synthetic data to enlarge the size of the training dataset. The above fact can help to mitigate the data availability limitation.
- **Transfer learning:** this can leverage pretrained DL models and fine-tune them to classify PQ events. In addition, it can help to reduce the required training data and improve the model’s overall performance.
- **Model explainability:** techniques such as feature visualization and attention mechanisms can increase the interpretability of DL models. This property can help to identify the most important features for classification, and provide insights into the underlying causes of PQ events.
- **Hybrid techniques:** the presence of noise in the PQ signal is considered one of the critical issues, and acts as a hindrance to accurately doing the feature detection and classification of PQ events. It is established that signal processing and artificial intelligence-based techniques with DL can provide a better method, having the potential to handle the issue due to their superiority in a noisy background.

The findings of different DL models and their comparative analysis are presented in Table 1.

Table 1. Comparison of deep-learning models.

Model	Advantages	Disadvantages	Time Complexity	Classification Accuracy	PQ Classifications Accuracy Range (%)	References
CNN	• Excellent for image and video data • Can learn spatial hierarchies • Parameter sharing reduces memory requirements	• Limited capability in capturing long-term dependencies • May require large amounts of training data	Varies depending on input size and network depth	High for image classification tasks	96.2–99.71	[10–17]
RNN	• Effective for sequence data and time-series analysis • Can capture long-term dependencies • Flexible input and output sizes	• Can be computationally expensive to train • Susceptible to vanishing/exploding gradient problems	Varies depending on sequence length and network size	High for sequential tasks	96.9–99.9	[18–28]
AE	• Unsupervised learning for data compression and reconstruction • Feature extraction and dimensionality reduction	• Limited performance on complex data • Requires labelled data for supervised pretraining	Depends on data dimensionality and network size	Moderate to high, depending on the dataset	94.5–98.4	[29–38]
GAN	• Can generate new data samples • Unsupervised learning without explicit labels • Can learn complex data distributions	• Training instability and mode collapse issues • Difficult to evaluate training progress	Depends on the generator and discriminator complexity	Varies depending on the task and dataset	96.2–98.6	[39–47]
DNN	• Powerful universal approximates • Can learn complex mappings • Suitable for large datasets	• Prone to overfitting without regularization techniques • High memory and computational requirements	Depends on the network architecture and depth	High with sufficient data and proper regularization	97.3–99.5	[48–54]

Table 1. Cont.

Model	Advantages	Disadvantages	Time Complexity	Classification Accuracy	PQ Classifications Accuracy Range (%)	References
DBN	- Effective for unsupervised pretraining - Stacked architecture for feature learning - Good at capturing hierarchical representations	- Training is time-consuming - Sensitive to hyperparameter settings	Depends on the network size and training iterations	Moderate to high, depending on the dataset	96.2–99.3	[55–61]
LSTM	- Handles long-term dependencies in sequential data - Captures context information effectively - Suitable for language processing tasks	- Computationally expensive - Can be prone to overfitting	Varies depending on sequence length and network size	High for sequential tasks, especially with long dependencies	92.3–99.3	[62–73]
DAE	- Unsupervised learning for feature extraction - Effective in de-noising and dimensionality reduction	- Requires labelled data for supervised fine-tuning - Can be sensitive to noise levels	Depends on data dimensionality and network size	Moderate to high, depending on the dataset	96.5–99.5	[74–82]
SOM	- Topological preservation of data structure - Efficient for visualization and clustering - Handles high-dimensional data well	- Convergence is slow for large datasets - May require tuning of hyperparameters	Depends on the network size and training iterations	Moderate, varies depending on the dataset and task	91.7–98.1	[83–91]
RMB	- Effective for unsupervised pretraining - Can learn hierarchical representation - Handles high-dimensional data well	- Training can be slow - Limited capability for deep architectures	Depends on the network size and training iterations	Moderate, varies depending on the dataset and task	93.7–99	[92–100]

5. Future Scopes

Many future scopes can be enumerated by looking at the research gaps and limitations of the existing and suggested approaches in recent times.

- Multimodal data-type power signals can provide additional information for PQ event classification. DL models can be extended to handle these types of data to enhance the total performance of the classification process.
- Real-time classification of PQ events is essential for quickly identifying and responding to events. DL models can be optimized for real-time classification using model compression and hardware acceleration techniques.
- Unsupervised learning techniques can automatically identify patterns and anomalies in PQ data. The above property can help to detect events that are not explicitly labeled, and offer a complete understanding of the overall condition of the power system.
- Robustness and generalization: DL models can be sensitive to changes in the data distribution, which can lead to a degradation in performance. Improving these models' robustness and generalization capabilities is an important area for future research.
- Privacy and security: power systems are critical infrastructures that must be protected against cyber-attacks. Deep-learning models can detect anomalies in the data that may indicate a security breach. However, it is also important to ensure that the models are secure in the model design, and that sensitive data are not leaked.
- Collaborative learning techniques can be applied for DL model training on distributed data sources while preserving data privacy. This technique can be particularly useful in cases where the data are sensitive or where the data are spread across multiple organizations.

- Human-in-the-Loop approaches can improve the interpretability and trustworthiness of DL models. These approaches involve incorporating human feedback and domain expertise into the training process to help identify and correct errors and biases in the model.
- Hybrid approaches, such as combining DL and expert systems, may bring better results to formulate a more automatic and explainable algorithm.
- Minimum data availability may degrade the DL model performance accuracy. Future research must focus on supervised and unsupervised learning to handle this issue in real time.
- Three phases: very few studies have been conducted in three-phase unbalanced systems integrating renewable energy sources, random and nonlinear loads, and power electronic devices.

6. Conclusions

This extensive review of DL application to PQ event detection reflects a few promising conclusions. Firstly, regarding efficiency, accuracy, and computational speed, DL-based approaches show significant promise for PQ disturbance detection and classification, addressing all limitations and issues in the existing methods. Secondly, DL approaches have not been exploited extensively in many smart grid applications as far as their capabilities might allow. They can be extended further to classify PQ disturbances, providing improved DL in accuracy, robustness, and interpretability. Thirdly, DL approaches for PQ analysis can be used in real time due to their inherent performance characteristics, such as faster computation, higher accuracy, and enhanced efficiency. Lastly, applying these techniques can significantly improve the PQ monitoring systems, ultimately enhancing the reliability and stability of smart grid systems, and creating a better protective and secure system. The ongoing research in this field promises to yield further improvements and advancements in DL-based PQ analysis, paving the way for a more sustainable and resilient energy future.

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